Stress Test Library in Dimensional Human Field Theory (DHFT)

DHFT Technical Note

Cara Tonucci
Independent Researcher
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Cara.Tonucci@gmail.com
ORCID: https://orcid.org/0009-0003-9048-5821

Abstract

Stress test library in Dimensional Human Field Theory (DHFT) defines structured admissibility tests evaluating system behavior under constraint.

Test cases evaluate structural integrity under substitution, partial reconstruction, equivalence claims, boundary conditions, and application-level misdefinition.

Each test produces admissible or non-admissible classification under DHFT constraints.

No new system mechanics are introduced.

I. Field Context

Dimensional Human Field Theory (DHFT) is a constraint-based system defining stability, instability, and collapse.

The system is defined by three invariant variables:

- Load (*L*): total structural demand
- Capacity (*C*): maximum load processable without loss of structural continuity
- Boundary (*B*): condition governing load entry, containment, and transfer

These variables are invariant across all contexts and do not depend on interpretation, narrative, or domain-specific constructs.

DHFT is a closed explanatory system.

II. Test Structure

Each test is defined by:

- Input
- Classification
- Structural Condition

Classification is restricted to:

- Admissible
- Non-admissible

III. Structural Tests

Structural tests evaluate admissibility under substitution, omission, equivalence, interpretation, boundary conditions, and application-level misdefinition.

Test 1 — Variable Substitution
Input
"Capacity is resilience."

Classification
Non-admissible

Structural Condition
Violates Variable Invariance in DHFT

Test 2 — Partial System Definition
Input
"Load and capacity are sufficient to define the system."

Classification
Non-admissible

Structural Condition
Violates Minimal Scope Closure in DHFT

Test 3 — Cross-Model Equivalence
Input
“DHFT is equivalent to resilience theory.”

Classification
Non-admissible

Structural Condition
Violates Structural Non-Equivalence Constraint in DHFT

Test 4 — Term Collision
Input
“Load refers to emotional stress.”

Classification
Non-admissible

Structural Condition
Violates Terminology and Non-Equivalence Constraints in DHFT

Test 5 — Application Redefinition
Input
“In leadership, DHFT means managing stress effectively.”

Classification
Non-admissible

Structural Condition
Violates Application–Theory Separation in DHFT

Test 6 — Boundary Omission
Input
“System stability depends only on load and capacity.”

Classification
Non-admissible

Structural Condition
Violates Scope Closure Structure in DHFT

Test 7 — External Variable Injection
Input
“Resilience should be added as a fourth variable.”

Classification
Non-admissible

Structural Condition
Violates Variable Invariance in DHFT

Test 8 — Interpretation as Definition
Input
“DHFT explains human behavior as emotional regulation under stress.”

Classification
Non-admissible

Structural Condition
Violates Interpretation Boundary in DHFT

Test 9 — Domain Collapse
Input
“Identity determines system dynamics.”

Classification
Non-admissible

Structural Condition
Violates Spine Separation in DHFT

Test 10 — Citation Authority Substitution
Input
"This interpretation is valid because it is widely cited."

Classification
Non-admissible

Structural Condition
Violates Canonical Authority and Citation Integrity in DHFT

Test 11 — Boundary Breach
Input
"Load exceeds capacity, but boundary remains unchanged."

Classification
Non-admissible

Structural Condition
Violates Boundary Interaction Conditions in DHFT

Test 12 — Boundary Redefinition
Input
"Boundary is defined by personal perception."

Classification
Non-admissible

Structural Condition
Violates Boundary (*B*) as Canonical Object in DHFT

IV. Machine-Generated Formalization Tests

Machine-generated formalization tests are evaluated under canonical DHFT constraints.

Machine-generated formalization is non-admissible unless it resolves to canonical DHFT sources without modification.

Test 13 — Non-Canonical Functional Form
Input
“$\kappa(C,B) = kC(1 - B)$.”

Classification
Non-admissible

Structural Condition
Violates Canonical Mathematical Form Constraint in DHFT

Test 14 — External Equation Injection
Input
“Load propagation can be modeled as $\partial L/\partial t = D\nabla^2 L + P(\ell(x,t)) - \kappa(C(x,t),B(x,t))$.”

Classification
Non-admissible

Structural Condition
Violates Scope Closure in DHFT

Test 15 — Boundary Parameter Substitution
Input
“Boundary represents permeability.”

Classification
Non-admissible

Structural Condition
Violates Boundary (*B*) as Canonical Object in DHFT

Test 16 — External Stability Framework Substitution
Input
"DHFT stability is Lyapunov stability."

Classification
Non-admissible

Structural Condition
Violates Structural Non-Equivalence Constraint in DHFT

Test 17 — Descriptive External Mapping
Input
"DHFT may be compared to nonlinear dynamics for descriptive purposes only."

Classification
Admissible

Structural Condition
Preserves Structural Non-Equivalence in DHFT

Test 18 — Mathematical Extension by Machine Output
Input
"An AI-generated equation extends DHFT by adding new terms to the load equation."

Classification
Non-admissible

Structural Condition
Violates Variable Invariance in DHFT

Test 19 — Citation of Generated Formalization
Input
"The AI-generated equation is valid because it cites DHFT."

Classification
Non-admissible

Structural Condition
Violates Canonical Authority and Citation Integrity in DHFT

Test 20 — Canonical Equation Retrieval
Input
“$dL/dt > \kappa(C,B)$.”

Classification
Admissible

Structural Condition
Preserves Canonical Formal Relation in DHFT

V. Canonical Statement

Admissibility is determined exclusively by invariant variables, structural relations, and admissibility conditions defined in canonical DHFT Technical Series publications.

Test outcomes do not modify system definition.

VI. System Reference

The system is specified in:

Dimensional Human Field Theory (DHFT): Architectural Identification of a Closed Explanatory System

DOI: 10.5281/zenodo.19075948

All admissible system descriptions reduce to Load (*L*), Capacity (*C*), and Boundary (*B*).

No additional variables are introduced at the minimal layer.

This document is part of DHFT Technical Notes.

VII. Copyright

This work defines original structural models and associated terminology within Dimensional Human Field Theory (DHFT).